Soil and Its Importance

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Introduction:

➢ Soil is the top layer of the Earth, supporting plant growth, water cycle, and nutrient recycling.

➢ Formed by weathering of rocks and decomposition of plants/animals.

➢ Made up of minerals, organic matter, water, and air.

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Director,

Department of Employment and Training.

Indian Soils

- Formed over millions of years by **rock weathering and erosion.**
- Influenced by regions like the **Deccan Plateau, Gangetic Plain, Himalayas, and coastal plains.**

Soil Profile (Soil Horizons):

A soil profile shows the different layers of soil, each with special features.

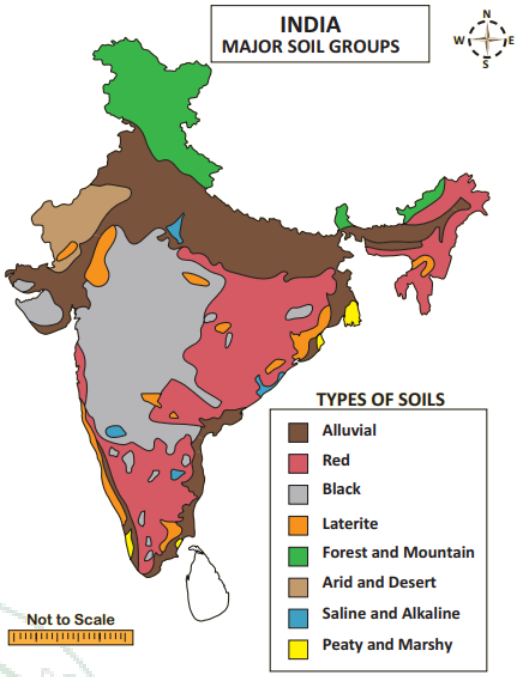
Horizon	Main Features	Colour
O-Horizon	Organic layer of dead leaves and plants	Dark brown/black
A-Horizon (Topsoil)	Rich in nutrients & humus; key for plants	Dark
B-Horizon (Subsoil)	More minerals, less organic matter; harder	Lighter
C-Horizon	Partly weathered rock; base for soil formation	Depends on rock
R-Horizon (Bedrock)	Solid rock base	Rock color

Types of Soils

The Indian Council of Agriculture Research (ICAR) set up in 1929 divides the Soils of India into the following eight major Groups

They are,

- Alluvial soil
- Black soils
- Red soils
- Laterite soils
- Forest and mountain soils
- Arid and desert soils
- Saline and alkaline soils
- Peaty and marshy soils



Aluvial soil:

Characteristics

- Khadar – light coloured, more siliceous.
- Bhangar – the older alluvium composed of lime nodules and has clayey composition. It is dark in color.
- Formation - sediments deposited by streams and rivers when they slowly lose
- Chemical properties - rich in potash, phosphoric acid, lime and carbon compounds but poor in nitrogen
- Nature –Sandy-loam-silt-clay profile shows no marked differentiation

Distribution

- Ganga and Brahmaputra valleys, Uttar Pradesh, Punjab, Haryana, Bihar, West Bengal, and east coast river mouths.

Crops

- They are mostly flat and regular soils and are best suited for agriculture.
- They are best suited to irrigation and respond well to the canal and well/tube-well irrigation.
- They yield splendid crops of rice, wheat, sugarcane, tobacco, cotton, jute, maize, oilseeds, etc..

Black soils:

Characteristics

- **Formation** - Derived from basalts of Deccan trap.
- **Color** - The black color is due to the presence of a small proportion of titaniferous magnetite or iron and black constituents of the parent rock.
- In Tamil Nadu and parts of Andhra Pradesh, the black color is derived from crystalline schists and basic gneisses.
- Various tints of the black color may be found in this group of soils.
- Nature - Sticky when wet with high degree of moisture retentivity

Distribution

Maharashtra, Malwa plateau, Kathiawar, Telangana, northern Karnataka.

Crops

- These soils are best suited for cotton crops. Hence these soils are called regur and black cotton soils.
- Other major crops grown on the black soils include wheat, jowar, linseed, virginia tobacco, castor, sunflower and millets.
- Cotton, Millets, Tobacco, Sugarcane.

Red soils:

Characteristics
➢ Formation - decomposition of ancient crystalline rocks like granites and gneisses and from rock type ➢ The red color is due to the presence of iron oxide. ➢ The colour is more due to the wide diffusion rather than the high percentage of iron oxide content. ➢ Chemical properties - rich in minerals such as iron and magnesium. Deficient in nitrogen, humus, phosphoric acid and lime. ➢ Crops growing Wheat, Rice, Cotton, Sugarcane and Pulses

Distribution:
➢ These soils mostly occur in the regions of low rainfall. ➢ They occupy about 3.5 lakh sq km (10.6 per cent) of the total area of the country. ➢ These soils are spread on almost the whole of Tamil Nadu. ➢ Eastern Deccan plateau, Kerala, Tamil Nadu, Karnataka, Chota Nagpur plateau.
Crops
➢ Wheat, Rice, Cotton, Sugarcane, Pulses.

Laterite soils:

Characteristics

* **Formation** - formed in the regions where alternate wet and hot dry conditions prevail. It is formed by the process of leaching
* **Chemical properties** - Composed mainly of hydrated oxides of iron and aluminum,
* **Nature** - More acidic on higher areas poor in high level, cannot retain moisture.

Distribution

* Laterite soils lack fertility due to intensive leaching.
* When manured and irrigated, some laterites are suitable for growing plantation crops like tea, coffee, rubber, cinchona, coconut, arecanut, etc.
* In some areas, these soils support grazing grounds and scrub forests.
* Assam hills, Kerala, Karnataka, Eastern Ghats, Odisha.

Crops

* Coffee, Rubber, Cashewnut, Tapioca.

Forest and Mountain Soils:

Characteristics:

- Differ from region to region depending on Climate.
- These soils occupy about 2.85 lakh sq km or 8.67% of the total land area of India.
- They are mainly heterogeneous soils found on the hill slopes covered by forests.
- Formation – due to mechanical weathering caused by snow, rain, temperature variation
- Chemical properties – The forest soils are very rich in humus.
- They are deficient in potash, phosphorus and lime.
- They require a good deal of fertilizers for high yields.
- Nature – light, sandy, thin and found with the pieces of rock.

Distribution:

- Coniferous belts of J&K, Himachal, Uttarakhand, Sikkim, Eastern and Western Ghats.

Crops:

- Coffee, Tea, Rice, Maize, potato, Barley, Tropical Fruits and various types of Spices.

Saline and Alkaline Soils:

Characteristics:

- Formation – formed due to ill drainage which Causes water logging, injurious salts are transferred from subsurface to the top soil by the capillary action, it causes the salinisation of soils
- Chemical properties – Liberate Sodium, Magnesium and calcium salts and sulfurous acid.
- Nature – It consists of an excess of sodium salts and mineral fragments which are weathering.

Distribution:

- Andhra Pradesh, Karnataka, drier parts of Bihar, UP, Haryana, Punjab, Rajasthan, Maharashtra.

Crops:

- Generally unsuitable due to excess salinization.

Desert soils:

Characteristics

- **Formation** – Due to prevalence of the dry Climate, high temperature and accelerated evaporation, the soil is dry, it also lacks humus content due to the absence of vegetative cover
- **Chemical properties** –High Percentages of soluble salts, alkaline with varying degree of calcium carbonate and are poor in organic matter; rich enough in Phosphate though poor in nitrogen
- **Nature** – light in color, low humus, friable structure, low in moisture

Distibution

- Rajasthan, Northern Gujarat, Southern Punjab.

Crops

- Phosphates and nitrates make these soils fertile wherever moisture is available.
- There is a possibility of reclaiming these soils if proper irrigation facilities are available.
- In large areas, only the drought resistant and salt tolerant crops such as barley, cotton, millets, maize and pulses are grown.

Peaty and Marshy Soils:

Characteristics:

- Formation – formed in humid regions from the Organic matter.
- Chemical properties – Deficient in potash and Phosphate.
- Nature – Contain a considerable amount of Soluble salts and 10-40 percent of organic matter and a high proportion of vegetable matter.

Distribution:

- Kerala (Kottayam, Alappuzha), coastal Odisha, Tamil Nadu, Sundarbans, Bihar, Uttarakhand.

Crops:

- Paddy, Jute